

Chemical Instabilities

Dissipative patterns are not exclusive of hydrodynamics but are also found in many other processes and particularly in chemistry. Some chemical reactions give rise to temporal and spatial variations of the mass concentrations of the active substances. In the course of time, some substances may undergo periodic oscillations whereas some spatial patterns may be formed in the reaction vessel. It should be stressed that these phenomena are underlying the activities of life and therefore far from being purely academic games. The central feature governing the appearance of self-organization in chemistry is the autocatalytic reaction: it is a reaction in which the products of some step take part to a subsequent step.

A representative example of chemical reaction exhibiting dissipative patterns is the BZ reaction, which is discussed later on. When chemical reactions are coupled with diffusion, as it occurs in spatially inhomogeneous mixtures, the corresponding spatio-temporal instabilities are designated as Turing's instabilities.

Maintenance of non-equilibrium states for a long period demands that chemical reagents are added to the system. But a result of such processes is that the concentrations of the various components are made very inhomogeneous and it is necessary to stir the mixture with the consequence that

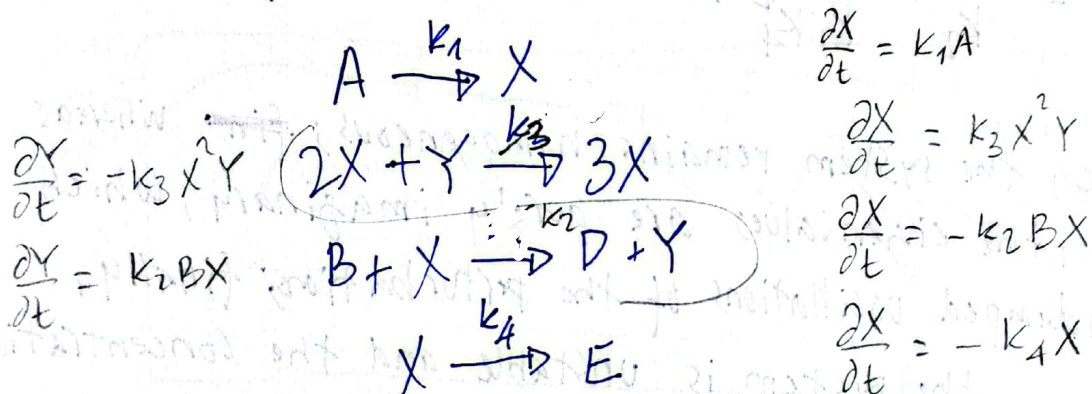
Spatial patterns will be destroyed. Only time oscillations can be observed in such stirred homogeneous reactions.

Oscillating Chemical Reactions

- In the early 1950s the Russian biochemist Boris Belousov was trying to create a test tube caricature of the Krebs cycle, a metabolic process that occurs in living cells.
- When he mixed citric acid and bromate ions in a solution of sulfuric acid, and in the presence of cerium catalyst, he observed to his astonishment that the mixture became yellow, then faded to colorless after about a minute, then returned to yellow to a minute later, then became colorless again, and continue to oscillate dozens of times before finally reaching equilibrium after about one hour.
- It was thought that all solutions of chemical ^{reactives} reagents must go monotonically to equilibrium, because of the laws of thermodynamics.
- Later on (1961) a graduate student Zhabotinsky confirmed to Belousov was right all along, and brought this work to light at an international conference in Prague in 1968 one of the few times that Western and Soviet scientists were allowed to meet.
- In 1980, BZ were awarded the Lenin Prize, the Soviet Union's highest medal, for their pioneering work on oscillating reactions.
- The BZ reaction is thought to involve more than twenty elementary reaction steps, but luckily many of them equilibrated rapidly, this allows the kinetics to be reduced to as few as three differential equations.

The Brusselator (Temporal Organization in Spatially Homogeneous System)

An appropriate modelling of the oscillatory behaviour of chemical reactions is provided by the **Brusselator** (Prigogine and Lefever, 1968; Nicolis and Prigogine, 1977) which consists in a simple and instructive example of autocatalytic scheme. The sequence of chemical reactions is the following:



the four steps are assumed to be irreversible which is achieved by taking all reverse reaction constants equal to zero; the global reaction of the above scheme is



- (1) The concentrations of the reactants A and B are maintained at a fixed and uniform non-equilibrium values, and the final products D and E are removed as soon as they are formed; the autocatalytic step is the second one, (2) which involves the intermediate species X and Y. It is assumed that the reactions take place under isothermal and well stirred, i.e. spatially homogeneous conditions (4).

According to previous results, we have seen that the

steady solutions are

$$X_s = \frac{k_1}{k_4} A$$

$$Y_s = \dots \frac{B}{A}$$

$$\frac{k_2 k_4}{k_1 k_3}$$

$$M = \begin{pmatrix} k_2 B - k_4 & k_3 X_s^2 \\ -k_2 B & -k_3 X_s^2 \end{pmatrix}$$

The stationary state becomes unstable (Re $\omega_k > 0$) when B is larger than a critical concentration given by

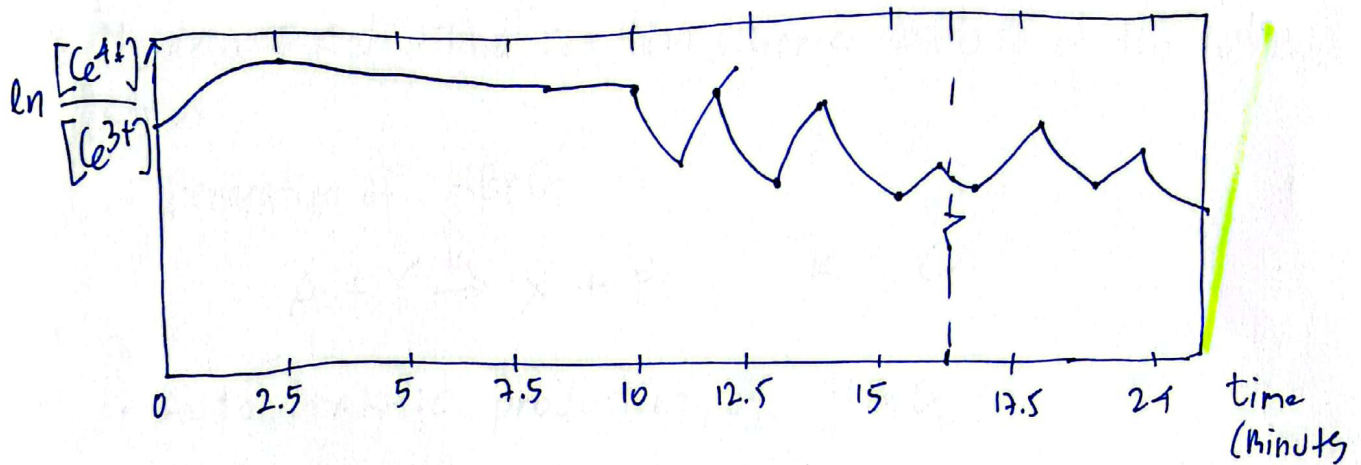
$$B_c = \frac{k_4}{k_2} + \frac{k_1^2 k_3}{k_2 k_4} A^2 \quad (1)$$

For $B < B_c$, the system remains homogeneous; ⁽²⁾ whereas at $B = B_c$, the eigenvalues are purely imaginary, which leads to undamped oscillations of the perturbations; finally, ⁽³⁾ for $B > B_c$, the system is unstable and the concentrations X and Y undergo periodic oscillations of the limit cycle type.

Such oscillations have indeed been detected in the BZ reaction, ^{en caso de no agitar} where fascinating geometric patterns as concentric circles and spirals propagating through the medium have been observed. (BZ) reaction is basically a catalytic oxidation by potassium bromate $KBrO_3$ of an organic compound such as malonic acid $CH_2(COOH)_2$ catalysed by cerium $Ce^{3+} - Ce^{4+}$ ion couple.
 Belousov-Zhabotinsky

After some transitory period, the oscillatory behaviour is evidenced by the variation of the concentration of the Ce^{3+} and Ce^{4+} ions. First, a blue colour indica-

ting an excess of Ce^{4+} is spreading into the mixture, a few minutes later the blue colour disappears and is replaced by a red one that indicates an excess of Ce^{3+} ions. The process goes on with a succession of blue, red, blue colours with a perfectly regular period, which to a certain sense constitutes a chemical clock.

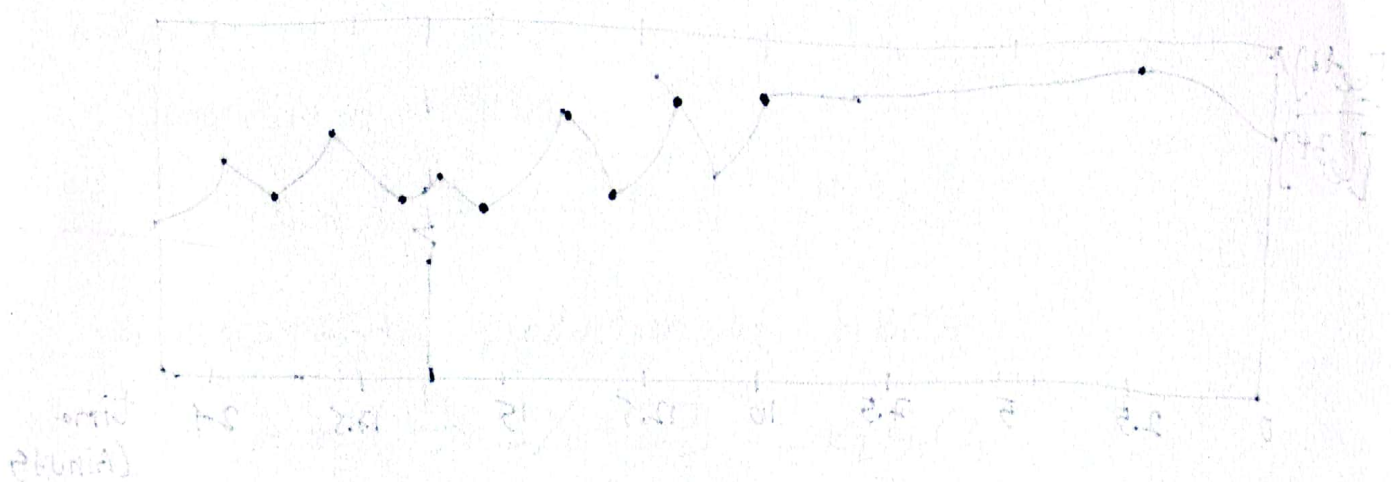


When the substances necessary for the reaction are exhausted, the oscillations will die out and the system goes back to its equilibrium state.

Spatial Organization in Spatially Heterogeneous Systems

In the absence of stirring, BZ reaction exhibits some non-trivial spatial patterns, which arise from the interplay of the chemical reaction and the diffusion process. When BZ reaction is performed in a long thin vertical tube so that the problem consists of a single spatial dimension, one observes a superposition of steady horizontal bands of different colours corresponding

to low and high concentration region. This structure is analogous to the Rayleigh-Bénard's and Taylor patterns.



Belousov-Zhabotinsky reaction (BZ)

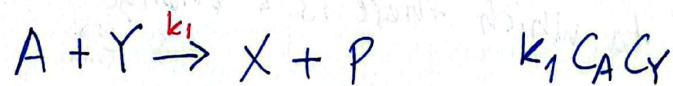
Kondepudi - Prigogine

⊗ Modelo FKN (Field, Körös, Noyes, J. Am. Chem. Soc., 94, 8649-8664 (1972))

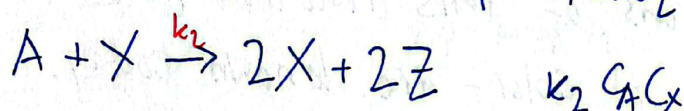
⊗ BZ is basically the catalytic oxidation of an organic compound, such as malonic acid $\text{CH}_2(\text{COOH})_2$.

⊗ Modelo FKN. The reaction scheme consists of the following steps:

• Generation of HBrO_2 :



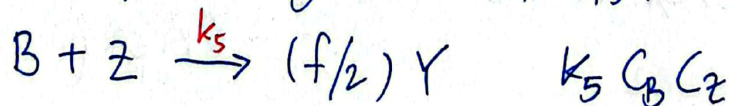
• Autocatalytic production of HBrO_2 :



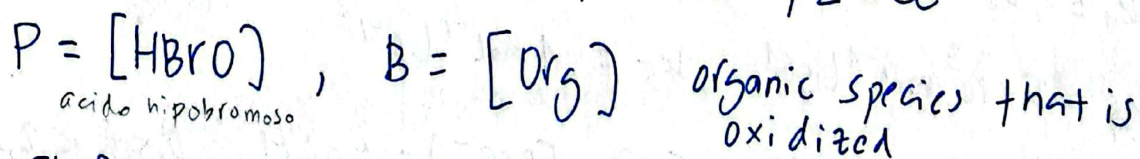
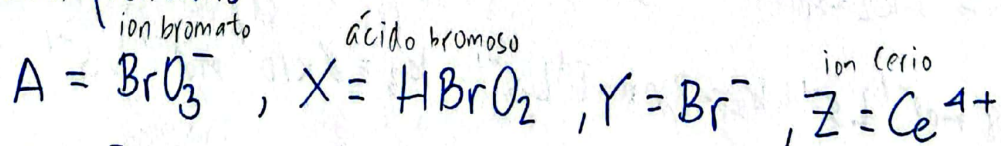
• Consumption of HBrO_2 :



• Oxidation of the organic reactants:



⊗ Identifications



$[\text{H}^+]$ is absorbed in the definition of the rate constant.

- ⊗ The reaction occurs in aqueous solution and is easily performed in a beaker by simply adding the following reactants in the concentrations shown:

$$[H^+] = 2,0 M, [CH_2(COOH)_2] = 0,28 M$$

$$[BrO_3^-] = 6,3 \times 10^{-2} M, [Ce^{4+}] = 2,0 \times 10^{-3} M$$

After an initial "induction" period, the oscillatory behavior can be seen in the variation of the concentration of the Ce^{4+} ion, due to which there is a change in color from colorless to yellow.

- ⊗ Many variations of this reaction (with more dramatic variations of color - are known today.

- ⊗ The corresponding rate equations:

$$\frac{dC_x}{dt} = k_1 C_A C_Y + k_2 C_A C_X - k_3 C_X C_Y - 2k_4 C_X^2$$

$$\frac{dC_Y}{dt} = -k_1 C_A C_Y - k_3 C_X C_Y + \frac{f}{2} k_5 C_B C_Z$$

$$\frac{dC_Z}{dt} = 2k_2 C_A C_X - k_5 C_B C_Z$$

$$k_1 = 1,28 \text{ mol}^{-1} \text{ L s}^{-1} \quad k_2 = 8,0 \text{ mol}^{-1} \text{ L s}^{-1} \quad k_3 = 8 \times 10^5 \text{ mol}^{-1} \text{ L s}^{-1}$$

$$k_4 = 2 \times 10^3 \text{ mol}^{-1} \text{ L s}^{-1} \quad k_5 = 1 \text{ mol}^{-1} \text{ L s}^{-1}$$

$$[B] = [O_3] = 0,02 M \quad [A] = [BrO_3^-] = 0,06 M \quad 0,5 < f < 2,4$$

Oscillations occur in this range